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3. Short Question Assignment solution:

Question 1: Jacobian Dimensionality

The Jacobian matrix will have exactly 128 rows and 512 columns (128 X 512).

Question 2: The Symmetric Superpower.

A) Clairaut's Theorem (or Schwarz's Theorem).

B) The spectral Theorem guarantees that the Hessian will always have strictly real eigenvalues and perfectly ~~Theorem guarantees that~~ orthogonal eigenvalues.

Question 3: Classifying the Critical Point

You are at a saddle point. The eigenvalues have mixed signs (positive 3.2, 0.4, 2.8 and negative -1.1). This means the surface curves upwards in three independt dimensions and downward in one.

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Question 4.

Second-order-optimization (Newton's Method)

Newton's Method requies computing, storing, and ~~int~~ inverting the entire Hessian matrix. For millions of parameters, H becomes to massive to fit in GPU memory, and inverting it ($O(N^3)$ complexity) would take centuries.

Question 5: ~~Jab~~ Jacobian vs. Gradient

The Gradient vector is strictly for scalar-valued function (many inputs → 1 output). The Jacobian matrix is for vector-valued functions (many inputs → many outputs).